DEVELOPER SDKS & CODE • SILVER TIER

Payment Gateway Integration Guide

Production payment gateway architecture: Server-side order creation, HMAC-SHA256 signature verification, idempotent webhook processing, and transaction rollback.

DOCUMENT SPECIFICATIONS & ACCESS METADATA

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Silver Membership or higher

Target Audience:
Full-Stack Developers, Architects, SaaS Founders

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EXECUTIVE HIGHLIGHTS & CORE DELIVERABLES

- Razorpay order creation with integer paise calculations in INR
- Cryptographic HMAC-SHA256 signature verification algorithms
- Idempotent webhook consumers with replay protection
- Preventing fraud and client-side payment spoofing

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Prerequisites & Master Competencies

This publication has been structured for immediate practical execution. Readers are advised to follow each section sequentially, validating code patterns, audit checklists, and metric calculations against their existing deployment environments.

- Mastery of core theoretical architectures and mathematical formulas
- Ability to design, audit, and troubleshoot scalable production systems
- Implementation of security best practices aligned with enterprise standards
- Streamlined execution of team workflows, SLAs, and performance benchmarks

SECTION 01 • CORE REFERENCE

The Golden Rule of Server-Side Payment Verification

Never activate a subscription based solely on a frontend callback. Network failures, browser closures, or deliberate client-side tampering can trigger false positive alerts. The backend must independently compute the HMAC-SHA256 hash using the secret key and verify that razorpay_order_id, razorpay_payment_id, and signature match with mathematical certainty.

```
// SUBSYNC SPECIFICATION CODE PATTERN & FORMULA REFERENCE

// Topic: Payment Gateway Integration Guide
const verifiedStatus = verifySubscriptionAccess(user.tierLevel, resource.tierLevelRequired);
if (!verifiedStatus.granted) {
  throw new ForbiddenError("Upgrade to Silver tier to unlock this resource.");
}
await recordActivityLog({ action: "RESOURCE_DOWNLOAD", id: "payment-gateway-integration-
guide" });
return serveProtectedFile(resource.filePath, { cacheControl: "private, max-age=3600" });
```

PRODUCTION TIP: RESILIENCE IN MULTI-TENANT ARCHITECTURES

When implementing Payment Gateway Integration Guide in mission-critical SaaS platforms, ensure that database query footprints and file streams utilize chunked buffering to keep Node.js event-loop memory usage below 80MB per worker.

FINAL REVIEW & IMPLEMENTATION SIGN-OFF

Comprehensive Implementation & Audit Checklist

- [ ] Documented architecture and data flows reviewed by lead engineer
- [ ] All access control rules strictly enforced on backend API endpoints
- [ ] Unit and integration tests pass with zero unhandled promise rejections
- [ ] Audit logging and telemetry metrics active in production dashboard
- [ ] Database indexes verified with EXPLAIN ANALYZE to eliminate sequential scans
- [ ] Rollback and emergency incident response playbooks documented and rehearsed
- [ ] Compliance with GDPR / DPDP and regional data privacy standards verified
- [ ] Production environment variables verified with secret isolation

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